

TASK 6: VISUALIZATIONS SUMMARY

CHARTS CREATED:

1. Box Plot - Monthly Charges by Churn:

- Churned customers paid \$13 more on the average section.
- Even the cheapest churned customers paid more than most loyal customers overall.

2. Violin Plot - Tenure by Churn:

- Half of the churned customers left within 10 months.
- Loyal customers spread across all tenures.

3. Correlation Chart - All Features vs Churn:

- Fiber optic internet is the first strongest churn driver.
- Electronic check is the second strongest churn driver.
- Long tenure = strongest churn preventer.
- Two year contract is the second strongest preventer.

4. Pair Plot - Key Variables Relationships:

- New high paying customers cluster together.
- Long term customers have high total charges, but low churn.
- Clear separation between the churned and stayed customers across all variables.

```
In [3]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv('Cleaned_Churn_Data.csv')

sns.set_style("whitegrid")
sns.set_palette("husl")

print("Shape:", df.shape)
```

Shape: (7043, 25)

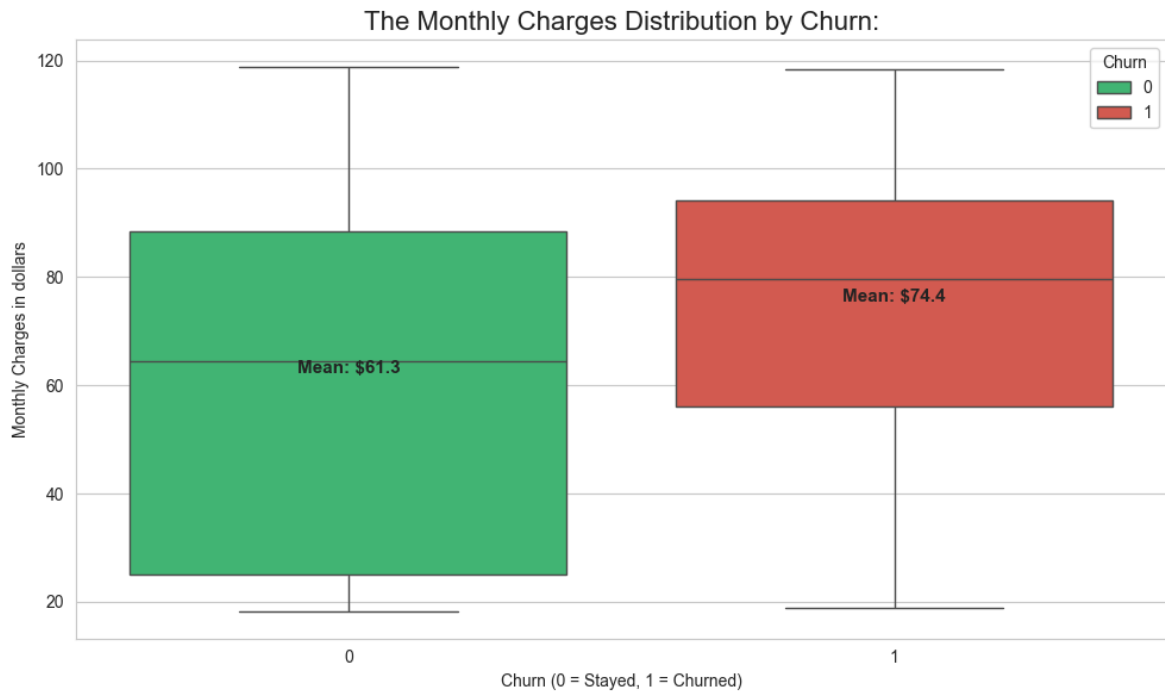
```
In [6]: plt.figure(figsize = (10, 6))
sns.boxplot(data = df, x = 'Churn', y = 'MonthlyCharges', hue = 'Churn',
            palette = ['#2ecc71',
                      '#e74c3c'])

plt.title('The Monthly Charges Distribution by Churn:', fontsize = 16)
plt.xlabel('Churn (0 = Stayed, 1 = Churned)')
plt.ylabel('Monthly Charges in dollars')

for i, group in enumerate([0, 1]):
    mean_val = df[df['Churn'] == group]['MonthlyCharges'].mean()
    plt.text(i, mean_val + 1, f'Mean: ${mean_val:.1f}', ha = 'center',
             fontsize = 11, fontweight = 'bold')

plt.tight_layout()
plt.show()
```

```
print("The Monthly Charges Statistics by Churn:")
print(df.groupby('Churn')['MonthlyCharges'].describe().round(2))
```



The Monthly Charges Statistics by Churn:

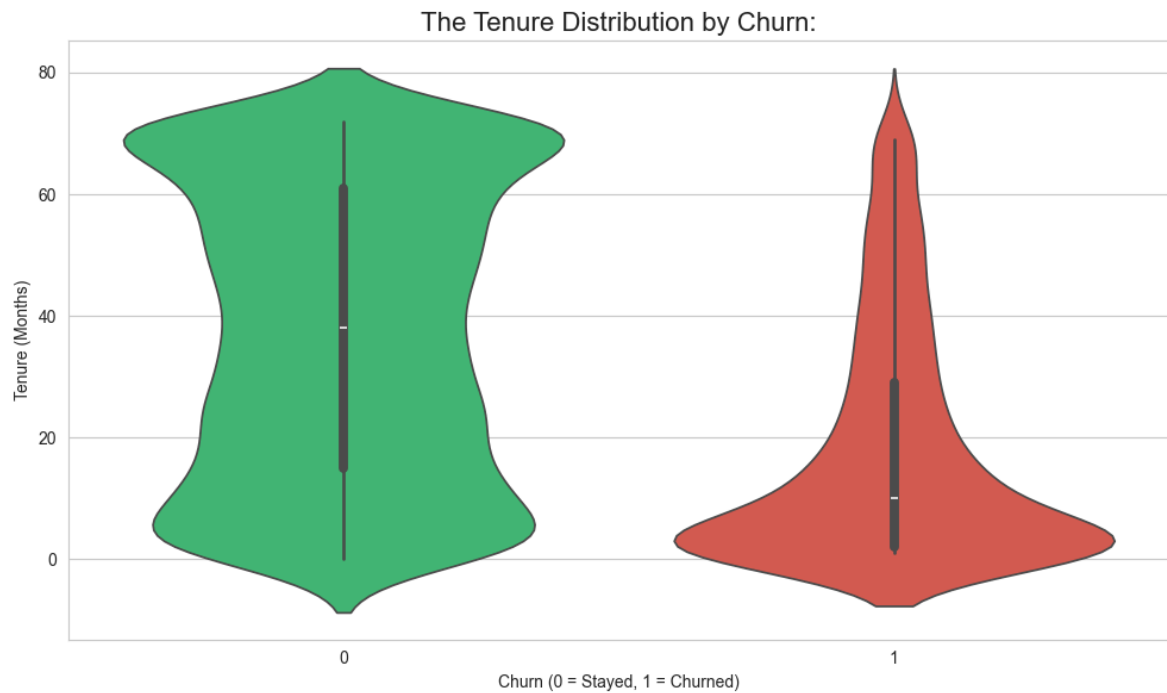
	count	mean	std	min	25%	50%	75%	max
Churn								
0	5174.0	61.27	31.09	18.25	25.10	64.43	88.4	118.75
1	1869.0	74.44	24.67	18.85	56.15	79.65	94.2	118.35

```
In [7]: plt.figure(figsize = (10, 6))
sns.violinplot(data = df, x = 'Churn', y = 'tenure', hue='Churn',
               palette = ['#2ecc71', '#e74c3c'], legend = False)

plt.title('The Tenure Distribution by Churn:', fontsize = 16)
plt.xlabel('Churn (0 = Stayed, 1 = Churned)')
plt.ylabel('Tenure (Months)')

plt.tight_layout()
plt.show()

print("The Tenure Statistics by Churn:")
print(df.groupby('Churn')['tenure'].describe().round(2))
```



The Tenure Statistics by Churn:

	count	mean	std	min	25%	50%	75%	max
Churn								
0	5174.0	37.57	24.11	0.0	15.0	38.0	61.0	72.0
1	1869.0	17.98	19.53	1.0	2.0	10.0	29.0	72.0

```
In [12]: plt.figure(figsize = (12, 8))

df_corr = df.select_dtypes(include = ['number'])

correlations = df_corr.corr()['Churn'].sort_values(ascending = False)

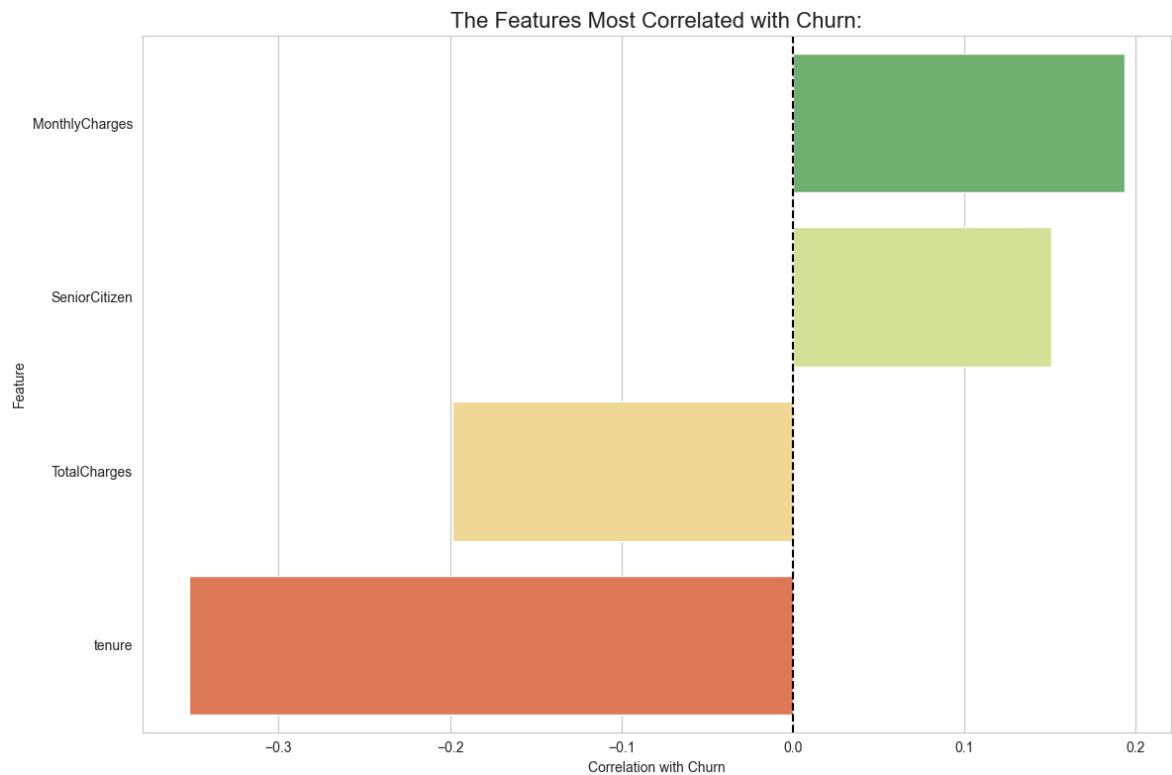
correlations = correlations.drop('Churn')

top_correlations = pd.concat([correlations.head(8), correlations.tail(5)])

sns.barplot(x = top_correlations.values, y = top_correlations.index,
            hue = top_correlations.index, palette = 'RdYlGn_r', legend = False)

plt.title('The Features Most Correlated with Churn:', fontsize = 16)
plt.xlabel('Correlation with Churn')
plt.ylabel('Feature')
plt.axvline(x = 0, color = 'black', linestyle = '--')
plt.tight_layout()
plt.show()

print("The Top factors that correlated with Churn:")
print(correlations.round(3))
```



The Top factors that correlated with Churn:

```
MonthlyCharges    0.193
SeniorCitizen     0.151
TotalCharges     -0.198
tenure           -0.352
Name: Churn, dtype: float64
```

```
In [6]: print(df.dtypes)
        print("These are the full columns")
```

```
customerID          str
SeniorCitizen      int64
tenure             int64
MultipleLines       str
OnlineSecurity      str
OnlineBackup        str
DeviceProtection    str
TechSupport         str
StreamingTV         str
StreamingMovies     str
MonthlyCharges     float64
TotalCharges       float64
Churn              int64
Contract_One year   bool
Contract_Two year   bool
PaymentMethod_Credit card (automatic) bool
PaymentMethod_Electronic check    bool
PaymentMethod_Mailed check        bool
InternetService_Fiber optic       bool
InternetService_No                 bool
gender_Male                        bool
Partner_Yes                        bool
Dependents_Yes                    bool
PhoneService_Yes                  bool
PaperlessBilling_Yes              bool
dtype: object
These are the full columns
```

```
In [9]: df_numeric = df.copy()

df_numeric = df_numeric.drop(columns = ['customerID'])

bool_cols = df_numeric.select_dtypes(include = ['bool']).columns
df_numeric[bool_cols] = df_numeric[bool_cols].astype(int)

str_cols = df_numeric.select_dtypes(include = ['object']).columns
print("The text columns being encoded:", str_cols.tolist())
df_numeric = pd.get_dummies(df_numeric, columns = str_cols, drop_first = True)

correlations = df_numeric.corr()['Churn'].sort_values(ascending = False)
correlations = correlations.drop('Churn')

print("\nAll correlations with Churn:")
print(correlations.round(3))
```

The text columns being encoded: ['MultipleLines', 'OnlineSecurity', 'OnlineBackup', 'DeviceProtection', 'TechSupport', 'StreamingTV', 'StreamingMovies']

All correlations with Churn:

InternetService_Fiber optic	0.308
PaymentMethod_Electronic check	0.302
MonthlyCharges	0.193
PaperlessBilling_Yes	0.192
SeniorCitizen	0.151
StreamingTV_Yes	0.063
StreamingMovies_Yes	0.061
MultipleLines_Yes	0.040
PhoneService_Yes	0.012
gender_Male	-0.009
MultipleLines_No phone service	-0.012
DeviceProtection_Yes	-0.066
OnlineBackup_Yes	-0.082
PaymentMethod_Mailed check	-0.092
PaymentMethod_Credit card (automatic)	-0.134
Partner_Yes	-0.150
Dependents_Yes	-0.164
TechSupport_Yes	-0.165
OnlineSecurity_Yes	-0.171
Contract_One year	-0.178
TotalCharges	-0.198
InternetService_No	-0.228
TechSupport_No internet service	-0.228
DeviceProtection_No internet service	-0.228
OnlineBackup_No internet service	-0.228
OnlineSecurity_No internet service	-0.228
StreamingTV_No internet service	-0.228
StreamingMovies_No internet service	-0.228
Contract_Two year	-0.302
tenure	-0.352

Name: Churn, dtype: float64

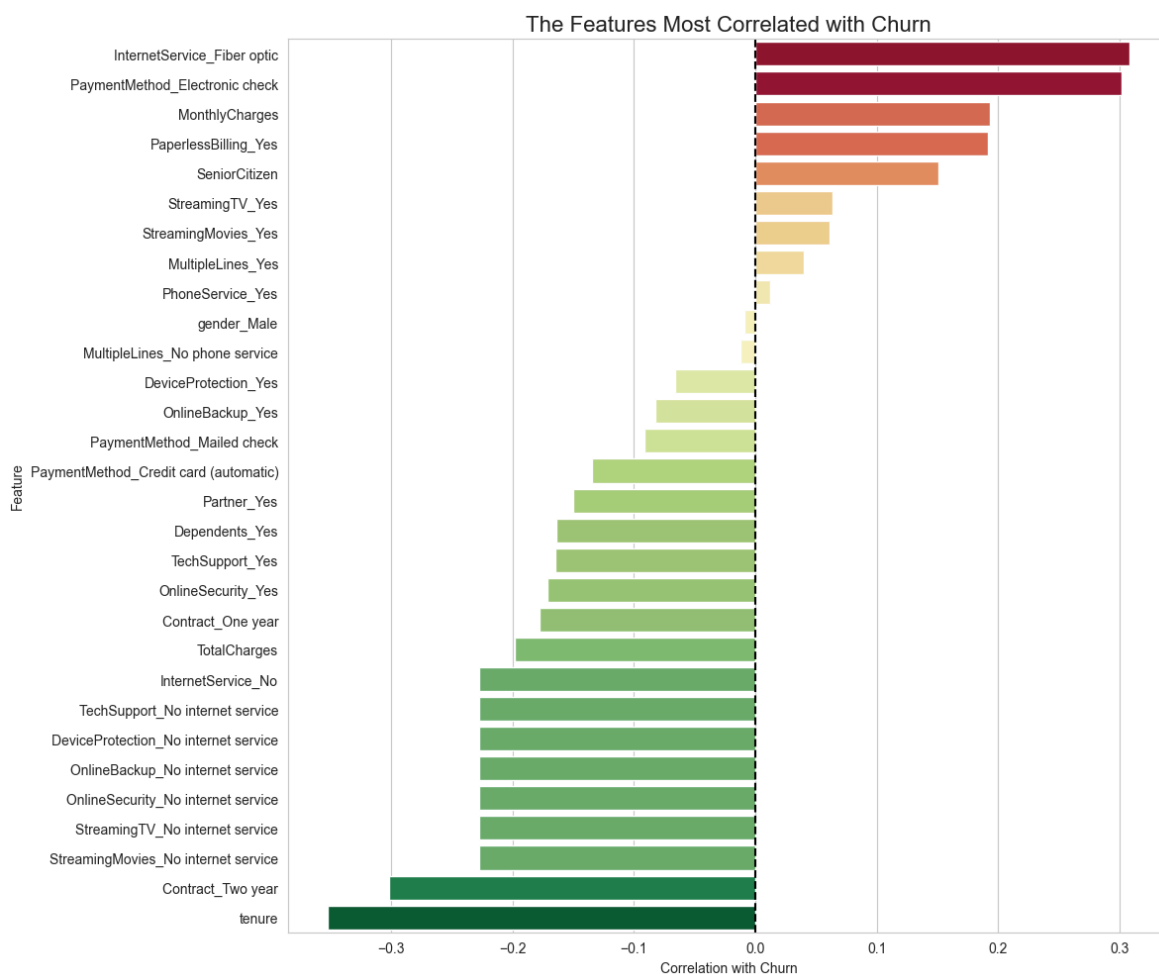
C:\Users\thend\AppData\Local\Temp\ipykernel_16592\3263328936.py:8: Pandas4Warning: For backward compatibility, 'str' dtypes are included by select_dtypes when 'object' dtype is specified. This behavior is deprecated and will be removed in a future version. Explicitly pass 'str' to 'include' to select them, or to 'exclude' to remove them and silence this warning.
See https://pandas.pydata.org/docs/user_guide/migration-3-strings.html#string-migration-select-dtypes for details on how to write code that works with pandas 2 and 3.

```
str_cols = df_numeric.select_dtypes(include = ['object']).columns
```

```
In [10]: plt.figure(figsize = (12, 10))

sns.barplot(x = correlations.values, y = correlations.index,
            hue = correlations.values,
            palette = 'RdYlGn_r',
            legend = False)

plt.title('The Features Most Correlated with Churn', fontsize=16)
plt.xlabel('Correlation with Churn')
plt.ylabel('Feature')
plt.axvline(x = 0, color = 'black', linestyle = '--', linewidth = 1.5)
plt.tight_layout()
plt.show()
```



```
In [11]: key_cols = ['tenure', 'MonthlyCharges', 'TotalCharges', 'Churn']

plt.figure()
pair_plot = sns.pairplot(df[key_cols], hue = 'Churn',
                        palette = ['#2ecc71', '#e74c3c'],
                        plot_kws = {'alpha': 0.5},
```

```
diag_kind = 'kde')

pair_plot.fig.suptitle('The Pair Plot of the Key Variables by Churn',
                        fontsize = 16, y = 1.02)

plt.show()
```

<Figure size 640x480 with 0 Axes>

